LAB EXPERIMENTS

Q2) Write a C program for monoalphabetic substitution cipher maps a plaintext alphabet to a ciphertext alphabet, so that each letter of the plaintext alphabet maps to a single unique letter of the ciphertext alphabet.

Aim

To write a C program to implement the **Monoalphabetic Substitution Cipher**, where each letter of the plaintext alphabet is replaced with a unique corresponding letter from a predefined ciphertext alphabet.

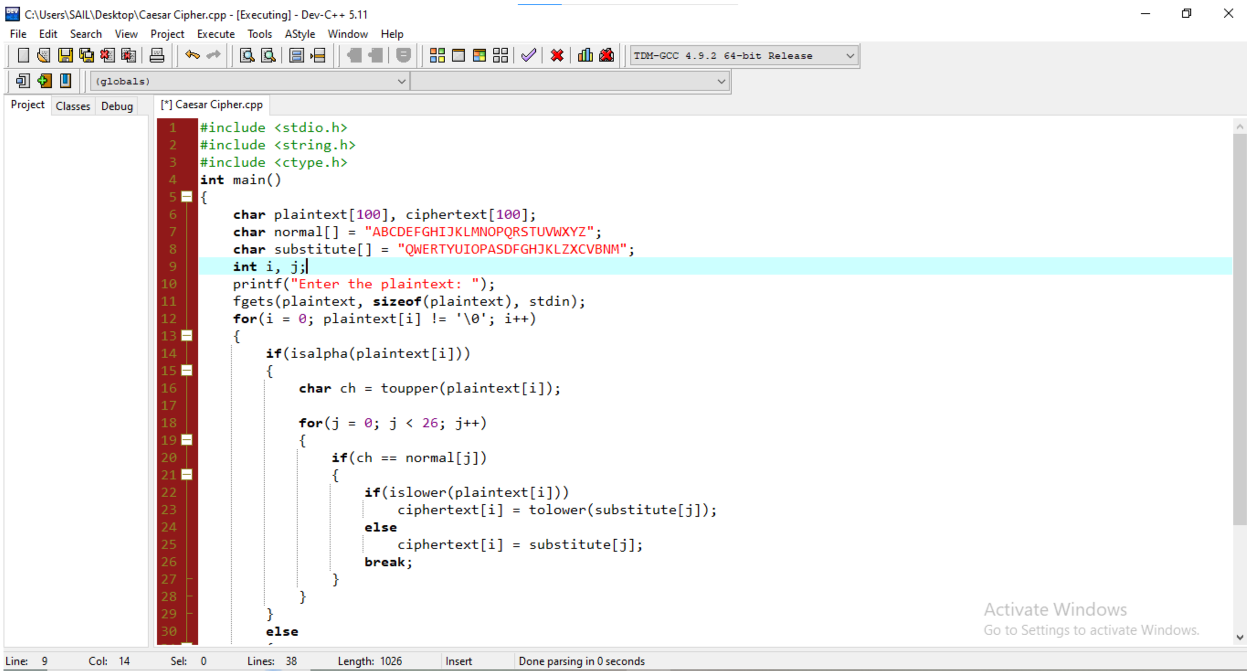
Algorithm

1. Start the program.
2. Declare arrays for the plaintext, ciphertext, normal alphabet, and substitution alphabet.
3. Initialize the normal alphabet as "ABCDEFGHIJKLMNOPQRSTUVWXYZ".
4. Initialize a substitution alphabet with a unique permutation of letters (e.g., "QWERTYUIOPASDFGHJKLZXCVBNM").
5. Read the plaintext message from the user.
6. Traverse each character of the plaintext.
7. If the character is an uppercase or lowercase alphabet, find its corresponding position in the normal alphabet and replace it with the mapped character from the substitution alphabet. Leave non-alphabetic characters unchanged.
8. Store the substituted characters in the ciphertext.
9. Display the encrypted ciphertext.
10. Stop the program.

Result

Thus, the C program to implement the **Monoalphabetic Substitution Cipher** was successfully executed, and the corresponding encrypted ciphertext was obtained.

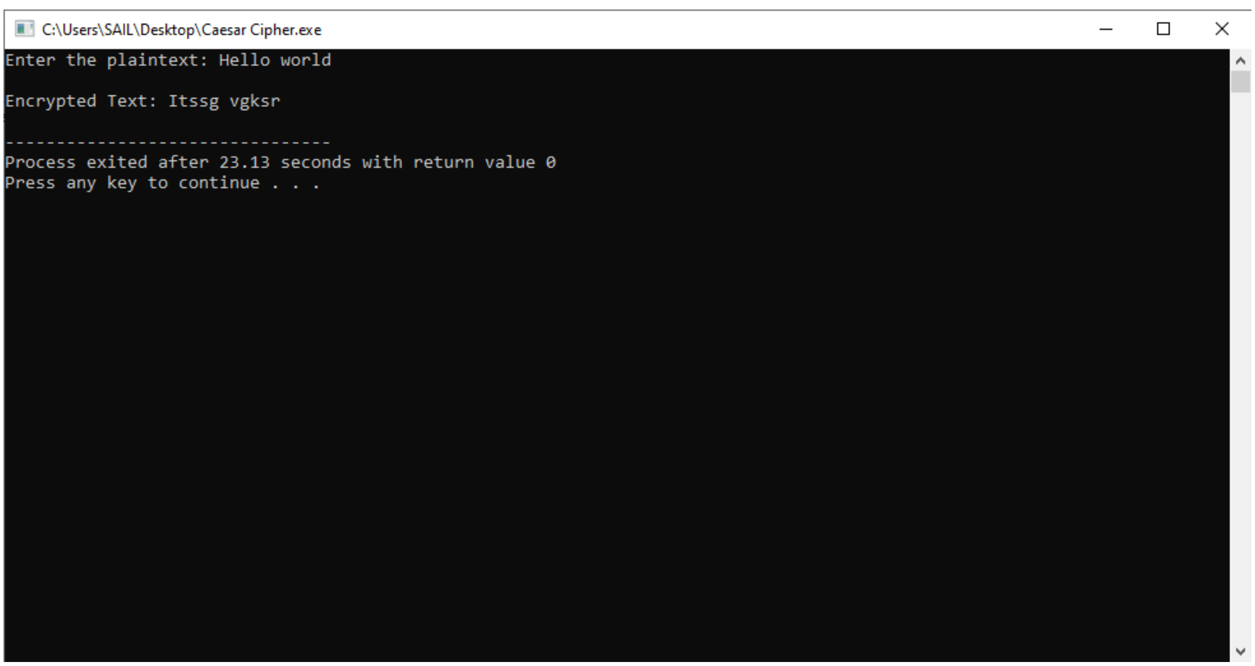
Code:



C:\Users\SAIL\Desktop\Caesar Cipher.cpp - [Executing] - Dev-C++ 5.11
File Edit Search View Project Execute Tools AStyle Window Help
TDM-GCC 4.9.2 64-bit Release
(globals)
Project Classes Debug
[*] Caesar Cipher.cpp
```
#include <stdio.h>
#include <string.h>
#include <ctype.h>
int main()
{
    char plaintext[100], ciphertext[100];
    char normal[] = "ABCDEFGHIJKLMNOPQRSTUVWXYZ";
    char substitute[] = "QWERTYUIOPASDFGHJKLZXCVBNM";
    int i, j;
    printf("Enter the plaintext: ");
    fgets(plaintext, sizeof(plaintext), stdin);
    for(i = 0; plaintext[i] != '\0'; i++)
    {
        if(isalpha(plaintext[i]))
        {
            char ch = toupper(plaintext[i]);

            for(j = 0; j < 26; j++)
            {
                if(ch == normal[j])
                {
                    if(islower(plaintext[i]))
                        ciphertext[i] = tolower(substitute[j]);
                    else
                        ciphertext[i] = substitute[j];
                    break;
                }
            }
        }
        else
```
Activate Windows
Go to Settings to activate Windows.
Line: 9 Col: 14 Sel: 0 Lines: 38 Length: 1026 Insert Done parsing in 0 seconds

Output



C:\Users\SAIL\Desktop\Caesar Cipher.exe
```
Enter the plaintext: Hello world

Encrypted Text: Itssg vgksr

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Process exited after 23.13 seconds with return value 0
Press any key to continue . . .
```